

Retail Business Performance & Profitability Analysis

Data Analyst Internship Project Report

Prepared by Kaviya	Project Retail Business Performance & Profitability Analysis	Dashboard Power BI
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1. Introduction

Retail businesses generate large volumes of transactional data covering sales, costs, products, inventory, customers and sales channels. Converting this information into clear business insights helps identify profitable product groups, understand sales patterns and support inventory-related decisions. This project analyzes a retail FMCG transaction dataset to evaluate business performance and profitability through data cleaning, exploratory analysis and interactive visualization.

The analysis focuses on category and brand profitability, revenue trends, store-format and channel performance, city-level sales, and the relationship between stock levels and units sold. An interactive Power BI dashboard was developed to present these findings in a concise business format.

2. Abstract

The project presents a data-driven analysis of retail FMCG sales and inventory information. The dataset was cleaned and transformed using Python and Pandas before being analyzed across multiple business dimensions. Key measures such as revenue, cost, profit, units sold and profit margin were calculated and used to build an interactive Power BI dashboard. The dashboard enables users to explore performance by category, brand, channel, store format and city, while also examining monthly revenue and profit trends and inventory behavior.

Project Objective

To analyze transactional retail data to identify profit-driving and profit-draining categories, understand sales and seasonal behavior, examine inventory indicators, and communicate the results through an interactive business intelligence dashboard.

■ 39.34M	■ 7.87M	~300K	20.02%
Total Revenue	Total Profit	Units Sold	Profit Margin

3. Tools Used

Tool	Purpose
Python & Pandas	Data loading, cleaning, transformation and aggregation.
Jupyter Notebook	Exploratory analysis and analytical workflow.
Power BI	Interactive dashboard development and visualization.
DAX	Measures for revenue, cost, profit, units sold and profit margin.

4. Steps Involved in Building the Project

- **Dataset preparation:** Imported the Indian FMCG retail sales and inventory dataset into Jupyter Notebook and created a working copy.
- **Data cleaning:** Converted invoice dates to a proper date format and handled missing customer-gender values using an 'Unknown' category.
- **Feature preparation:** Created a Year-Month field for time-based analysis and a Low-Stock Flag by comparing stock on hand with reorder level.
- **Profitability analysis:** Aggregated revenue, cost, profit and units sold by category and brand, and calculated profit margin percentages.
- **Sales analysis:** Examined monthly performance, city-level sales, store-format performance and channel contribution.
- **Inventory analysis:** Compared stock on hand, reorder level, lead time and units sold to identify inventory-related patterns.
- **Power BI development:** Imported the cleaned dataset, created DAX measures and built KPI cards, trend charts, category/brand visuals, channel analysis, store-format analysis, map, scatter plot, waterfall and treemap.
- **Interactivity:** Added Category, Store Format and City slicers for dynamic filtering.

5. Key Findings

- The analysis generated approximately **₹39.34 million in revenue** and **₹7.87 million in profit**.
- Approximately **300,000 units** were sold across the analyzed retail categories.
- The overall calculated **profit margin is 20.02%**.
- Online, Offline and Omnichannel sales can be compared directly through the channel visualization.
- Express, Hyper and Super store formats show closely distributed revenue in the dashboard.
- Category-level profit and inventory visuals support identification of areas requiring closer profitability and replenishment attention.

6. Dashboard Output

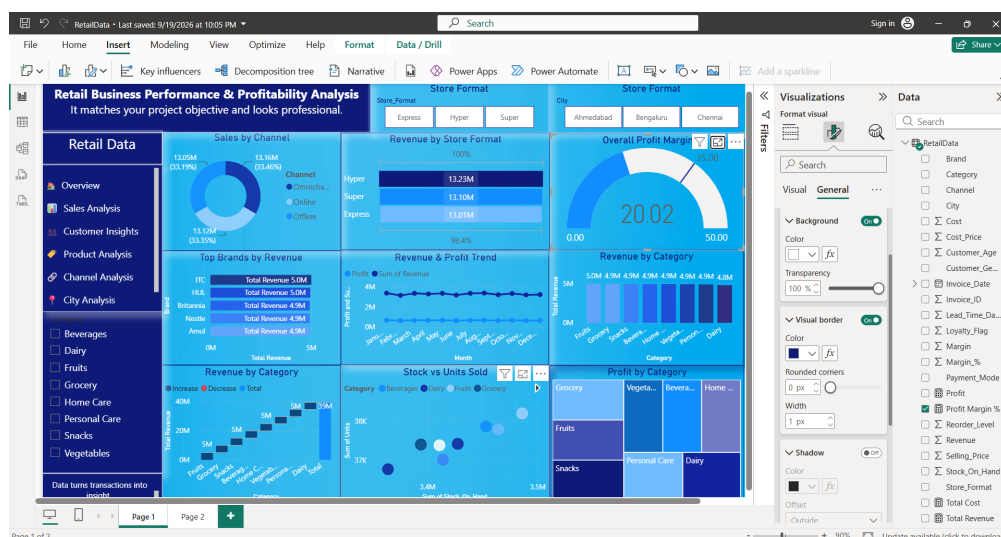


Figure 1. Interactive Power BI dashboard for retail business performance and profitability analysis.

7. Conclusion

This project demonstrates how transactional retail data can be transformed into useful business information through a structured data-analysis workflow. Python and Pandas were used for data preparation and analysis, while Power BI was used to communicate the results through an interactive dashboard. The final analysis provides

a consolidated view of revenue, profit, sales channels, store formats, products and inventory indicators, supporting clearer performance monitoring and data-driven decision-making.

Dataset: Indian FMCG Retail Sales Customer Inventory (2024) | **Project Type:** Data Analytics / Business Intelligence